作业来自

梁昆淼《力学》 第 4 版 高等教育出版社

Johns Analytical Mechanics 2nd ed Oxford University Press

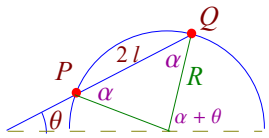
例如： L01.08 表示梁昆淼书第 01 章第 08 题； J02.05 Johns 书第 02 章第 05 题

J03.07† 表示 Johns 书第 03 章第 07 题，带 † 表示仅要求荣誉班同学完成

Homework 05, Week 07 2024.10.16 Due 2024.10.23 L02.05, L02.09, J03 .07†, J03.08, J03.09, J03.10, 补充题

★ L02.05

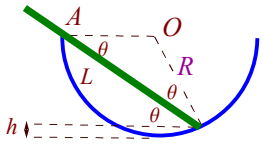
光滑钢丝弯成半径为 R 的半圆竖直放置，钢丝上穿着两个质点，
其重量分别为 P 和 Q，两质点由 长度为 $2l$ 的不可伸长
轻绳连起来，求平衡时绳与水平面的夹角，
如图 （图参照教材 p442）



Solution

★ L02.09

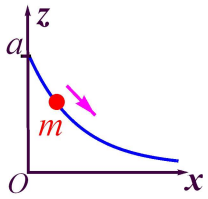
光滑半球碗是固定的，匀质杆长 L 斜靠在碗边，
杆的下端为碗所承托，
上端则伸出碗外，试求杆的平衡位置。
在怎样条件下才有可能平衡？



Solution

★ J03.08

A bead of mass m slides without friction on a rigid wire that lies in the xoz plane and has the shape $z = a\,e^{-\gamma x}$, where γ is some given constant.
Gravity acts downwards, with $\vec{g} = -g\,\vec{e}_z = -g\,\vec{e}_3$.



(a) Write the full Lagrangian for this problem, and write the equations for the two holonomic constraints (几何约束), first that the mass is confined to the plane $y = 0$, and second that it is confined to the surface $z = a\,e^{-\gamma x}$.

(b) Write the three Lagange equations, intoducing the Lagrangian multipliers λ_1 and λ_2 as appropriate.

(c) Use the constraints to write a reduced Lagrangian $\tilde{L}(x, \dot{x}, t)$, with x serving as the single free coordinate.

Derive the reduced generalized energy from this reduced Lagrangian, and use it to find an expression for $\dot{x}^2$ as a function of x. (Assume that

the mass is released from rest at the point $x = 0$.) Also use
the reduced Lagrange equation to find an expression for $\ddot{x}$ as a function of x and $\dot{x}$

(d) Write an expression for the Cartesian vector force of the constraint $\vec{F}^{(\text{cons})}$ acting on the bead, expressing

it as a function of x only. Check the limit of $\vec{F}^{(\text{cons})}$ as $x \to \infty$. Is it reasonable?

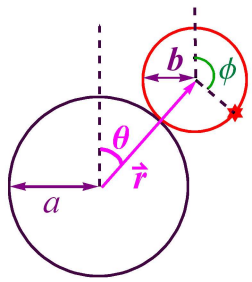
Solution

★ J03.09

A fixed, right circular cylinder (first cylinder) of radius a lies on its side, with its symmetry axis horizontal. A hollow right circular cylinder (second cylinder) of radius b and mass m is free to roll without slipping on the first one. Assume that its symmetry axis ramains aligned with that of the first cylinder. The full Lagragian for the second cylinder 's motion is

$$L = \frac{1}{2} m\left(\dot{r}^2 + r^2 \dot{\theta}^2\right) + \frac{1}{2} m b^2 \dot{\phi}^2 - m g r \cos\theta \qquad (1)$$

where r is the distance between the axes of the two cylinders, and θ and ϕ are angles shown in the figure.



Notice that ϕ is the angle between vertical and a mark ✱ inscribed on the face of the second cylinder.

(a) Write the two constraint functions, G_1 expressing the constraint that second cylinder is in contact with the first one, and G_2 expressing the condition of rolling without slipping. Assume that $\phi = 0$ when $\theta = 0$

(b) Write a (completely) reduced Lagragian $\tilde{L}\left(\theta, \dot{\theta}, t\right)$ and use the reduced generalized energy theorem to express $\dot{\theta}$ as a function of θ. Assume the second cylinder to be initially at rest, and at a very small distance to the right of $\theta = 0$.

(c) Use the Lagrangian given in Eq. (1) to write the three Lagrange equations, introducing Lagrange multipliers as appropriate. Find the generalized force of constraint $Q_r^{(\text{cons})}$ for the r variable and use it to find the angle θ_c at which the rolling cylinder will lose contact with the fixed one.

Solution

★ J03.10

Consider a bead of mass m sliding without friction on a wire in the x-o-z plane bent into shape of an inverted parabola $z = a - b\,x^2$. At time zero, the bead is at $x = 0$, $z = a$ with a velocity $\vec{v}_0 = v_0\,\vec{e}_x = v_0\,\vec{e}_1$. Constants a, b, v_0 are all greater that zero. Gravity acts in the negative z direction.

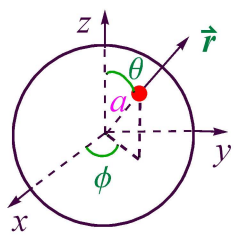
(a) Write an expression for the vector constraint force as a function of x only.

(b) Find the value of v_0 as a function of g, a, b which will make the constraint force exactly zero for all time $t > 0$.

Solution

★ J03 .07† （荣誉课程同学完成）

A single mass m slides without friction on the outside surface of a sphere of radius a. Suppose that at $t = 0$, it has spherical polar coordinates $\theta_0 > 0$, $\phi_0 = 0$ and generalized velocities $\dot{\theta}_0 = 0$ and $\dot{\phi}_0 > 0$.



(a) Using spherical polar coordinates, write both the full Lagrangian and the reduced Lagrangian for this porblem.

"full" 和 "reduced" 分别指拉格朗日中没有消去和消去受限坐标 $q^{(b)}$, $\dot{q}^{(b)}$ 情况。

(b) Write the Lagrange equations for both the full and the reduced Lagrangian

(c) Use the reduced Lagrangian $\tilde{L}$ to write the reduced generalized energy $\tilde{H}_q$

(d) The mass will leave the surface of the sphere at the instant at which the normal force of the constraint becomes negative. Write an expression for the angle $\theta_{\max}$ at which the mass leaves the sphere.

(e) Define the parameter γ by $a\,\dot{\phi}_0^2 = \gamma\,g$, where g is the acceleration of gravity. Express $\theta_{\max}$ as a function of a, g, γ, θ_0 only. In the case with $\gamma = 0$, find $\theta_{\max}$ in the limit of $\theta_0 \to 0$.

(f) With $\theta_0 = 10\,°$, find the numerical value of $\theta_{\max}$ for the case in which $\gamma = 0$. How big would γ have to be in order to reduce $\theta_{\max}$ to $45\,°$.

Solution

★ 瑞利耗散势

质点受到正比于速度的阻尼力 $\vec{F}_i = -k_i \vec{v}_i$ (常数 $k_i \geq 0$)，其中 $\vec{v}_i$ 为相对于阻尼背景的速度。

这种力可视为非保守力写在拉格朗日方程的右边，也可以引入瑞利耗散势 $\mathcal{F}_i = \frac{1}{2} k_i v_i^2$ 描述，

阻尼力 $\vec{F}_i = -\nabla_{\vec{v}_i} \mathcal{F} = -\frac{\partial \mathcal{F}}{\partial v_{ix}} \vec{e}_x - \frac{\partial \mathcal{F}}{\partial v_{iy}} \vec{e}_y - \frac{\partial \mathcal{F}}{\partial v_{iz}} \vec{e}_z$。此外质点还受到势为 $V(\vec{r})$ 的保守力的作用

(a) 试在 s-系中写出以非保守力形式呈现的拉格朗日方程；

(b) 试在 s-系中写出以瑞利耗散势形式呈现的拉格朗日方程；

(c) 试证明在任意广义坐标 q-系中，修正的拉格朗日方程为 $\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_\alpha}\right) - \frac{\partial L}{\partial q_\alpha} + \frac{\partial \mathcal{F}}{\partial \dot{q}_\alpha} = 0$；

注意耗散势并未归入拉格朗日量，只是在拉格朗日方程中多出一项。

(d) 引入瑞利耗散势的优点是拉格朗日方程具有协变性，试利用瑞利耗散势写出受到正比于速度的阻尼力的单摆的拉格朗日方程。

Solution

End